Enhancing Student Self-Efficacy and Interest in Microelectronics through Immersive Virtual Reality in an Informal Learning Environment

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***Abstract*—This study investigates iVRLab, an immersive virtual reality (VR) microfabrication training system, and its influence on college students' self-efficacy and interest in microelectronics during a one-hour workshop held within a 1.5-day training camp. iVRLab simulates photolithography cleanroom operations, providing hands-on virtual experiences that merge theory with practice. A pre–post analysis revealed a significant rise in self-efficacy (M = 4.11 to 4.37, p = 0.0066), demonstrating the system's effectiveness in building confidence. Although interest only showed slight gains, high baseline scores (3.6–3.7 on a 4-point scale) indicate a ceiling effect. Correlations among engagement, embodiment (r = 0.91, p < 0.001), and immersion (r = 0.60, p < 0.01) underscore the value of active, embodied VR learning. Qualitative feedback further highlighted iVRLab's immersive realism and its capacity to spark curiosity about microfabrication careers. These findings suggest that VR training can effectively boost self-efficacy in microelectronics, reinforcing students' enthusiasm in an informal learning setting and indicating the potential broader adoption across STEM fields.**

Keywords—Virtual Reality, Microelectronics Education, Self-Efficacy, STEM Education, Interest in STEM

I. Introduction

Microelectronics is transforming technology, from wearable devices to advanced brain-machine interfaces and AI systems. Yet, despite its importance, expertise remains scarce. The semiconductor industry, projected to surpass $1 trillion by 2030 [1], faces a severe skills shortage, with 20,000 unfilled U.S. positions in 2022 and an additional 50,000 needed within five years [2]. Governments, academia, and industry must collaborate on robust education and workforce development programs [3].

Initiatives like those by the National Science Foundation aim to strengthen microelectronics education from K-12 to post-graduate levels [4]. However, only 15% of U.S. students met STEM benchmarks in 2023, down from 20% just a few years ago [5]. This "leaky STEM pipeline" arises from perceived subject difficulty, limited early exposure, and scarce hands-on learning opportunities [6], [7]. Microelectronics, a key STEM component, often faces even greater challenges, underscoring the need for early engagement and clearer career pathways.

Virtual Reality (VR) is transforming microelectronics education by seamlessly integrating theory with practice. Traditional labs are often expensive and limited, restricting student access. VR bypasses these barriers through realistic simulations, allowing learners to perform complex tasks without risk or high costs.

Immersive VR fosters embodied learning via hands-on activities that replicate real-world operations. Studies show it boosts understanding, retention by enabling safe, repeated practice [8], [9]. Additionally, VR significantly improves motivation, engagement, and perceived learning outcomes compared to traditional systems, creating a sense of presence and emotional connection that makes abstract concepts more accessible and engaging [10]. These findings demonstrate VR's potential as a powerful tool for microelectronics education.

Enhancing self-efficacy and maintaining long-term interest in STEM are crucial for bridging workforce gaps in science and engineering. VR systems offer risk-free, hands-on learning environments that not only build confidence but also engage learners through interactive tasks. Research demonstrates that VR effectively boosts self-efficacy and enhances engagement, solidifying its role as an indispensable tool in STEM education [11], [12]. Moreover, immersive VR experiences spark curiosity and deepen connections to STEM fields, ensuring sustained interest over time [13]. Meanwhile, teacher and student acceptance of VR is vital for its adoption and impact on self-efficacy. Barrett et al. [14] found that perceived usefulness, ease of use, and immersive features drive learners' intent to use VR. Mystakidis et al. [15] showed teachers favor VR, especially with game-based elements like virtual escape rooms,

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for boosting motivation and outcomes. Enhancing acceptance is key to integrating VR in STEM education.

In conclusion, VR holds significant potential to revolutionize microelectronics education by enhancing engagement, self-efficacy, and interest through immersive, hands-on learning. As the semiconductor industry expands, integrating VR tools into education will be crucial for developing a skilled, confident workforce. However, limited research exists on VR's impact on college students' self-efficacy and sustained interest in microelectronics, necessitating further exploration.

Our research team developed the iVRLab [16], an immersive VR training system simulating key microelectronics fabrication processes to enhance undergraduate semiconductor education. By bridging theoretical knowledge and practical application, iVRLab fosters both technical skills and affective development. This study evaluates iVRLab's impact on students' affective factors and learning experiences in an informal learning setting, investigating the following research questions:

RQ1: How does the iVRLab intervention impact students' self-efficacy in microelectronics before and after participation?

RQ2: How does the iVRLab intervention impact students' interest in microelectronics before and after participation?

RQ3: How do students perceive other aspects of their learning experiences, such as engagement, immersion, and cognitive load, during the iVRLab intervention?

II. METHODOLOGY

A. *Theoretical framework*

The iVRLab's design draws on Mayer's Cognitive Theory of Multimedia Learning (CTML) [17], which highlights the effectiveness of dual-channel processing, limited capacity, and active integration. The iVRLab aligns with these principles, offering immersive, interactive settings that enhance self-efficacy, retention, and problem-solving through simulated cleanroom processes, with its design detailed clearly in [16].

B. *Participants*

This study involved 27 participants. Participants ranged in age from 18 to over 50, specifically: 16.7% were 18–19, 41.7% were 20–23, 20.8% were 24–29, 16.7% were 30–39, and 4.2% were 50 or older. Their educational backgrounds ranged from high school diplomas (41.7%) to associate (33.3%), bachelor's (16.7%), and master's (4.2%) degrees. Fields of study included Electrical Engineering (58.3%), Computer Science (20.8%), Computer Engineering (12.5%), Finance and Economics (4.2%), and Industrial Engineering (4.2%). Of these participants, 24 completed the demographic survey, 26 completed both pre- and post-surveys, and 7 participated in follow-up interviews.

C. *iVRLab*

The iVRLab was developed to immerse students in photolithography cleanroom operations, replicating industry-aligned processes such as wafer cleaning and alignment/exposure [16]. Based on real lab observations, it connects theory and practice through interactive, scaffolded training with real-time feedback (see Figure 1). The training was grounded in research-based pedagogical strategies, including sequenced tasks with increasing complexity, immediate corrective feedback, and learner-controlled pacing. These practices promoted active engagement and self-directed learning. Multiple rounds of pilot tests have shown iVRLab's effectiveness in enhancing skills and engagement in microelectronics education [8], [16].

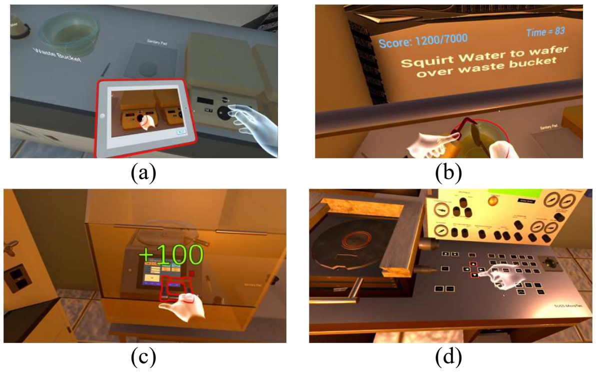

Fig. 1. iVRLab features: (a) Tasks-driven tutorials. (b) Canvas instructions. (c) Points for correct operations. (d) Object highlights.

D. *Research procedure*

The iVRLab activity was part of a 1.5-day microelectronics weekend camp held in Fall 2024 at a land-grant university in the American Midwest, approved by the Institutional Review Board (IRB). Conducted on a Saturday afternoon in a spatial computing lab equipped with Meta Quest 2 headsets, it offered each participant a hands-on microfabrication experience. 27 participants were divided into three groups, each with 7–10 students (see Figure 2). Facilitators assisted with technical issues as needed. The VR system was single-user; each student interacted with individually with the simulation. The one-hour individual training began with a brief VR warm-up to familiarize participants with the equipment, followed by a single training session where students performed photolithography tasks with step-by-step guidance and interactive feedback. A subset of participants later volunteered for interviews, providing qualitative insights into their experiences.

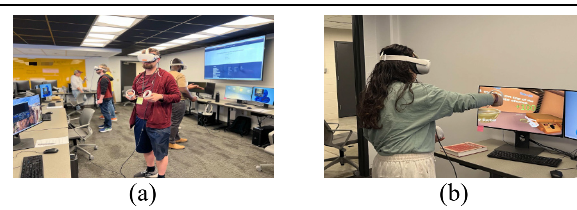

Fig. 2. iVRLab intervention (a) Participants engaged in a training session. (b) Individual spatial setup.

E. *Data collection*

Data were collected through pre- and post-surveys and semi-structured interviews. Surveys measured self-efficacy

[18], interest in microelectronics [19], cognitive load [20], [21], engagement [22], embodiment [23], [24], and immersion [25]. Self-efficacy and interest were assessed before and after the intervention, while the other constructs were evaluated in the post-survey only. All items were adapted for microelectronics. Interest used a 4-point scale, while all other constructs used a 5-point scale.

The survey instrument for interest in microelectronics [19], included two sub-constructs: Ideas about Careers, which assessed STEM motivation and career aspirations (e.g., "*I would like to work in a field related to microelectronics*"), and Ideas about Microelectronics, which evaluated curiosity and engagement (e.g., "*Microelectronics discoveries improve the quality of our life*"). Other constructs included items specific to microelectronics, such as "*I'm confident I can use the basic concepts taught in the iVRLab to operate the machines in the actual cleanroom*" for self-efficacy, and "*I felt the task very much mentally demanding*" for cognitive load. Engagement, embodiment, and immersion were captured through statements like "*I was engaged in this activity*," "*I felt like my virtual hands/arms were actually part of myself*," and "*It was as if I could interact with the VR world as if I was in the real world*," respectively. Instruments demonstrated strong reliability, with Cronbach's Alpha values ranging from 0.788 to 0.945.

Semi-structured interviews supplemented the survey data. Conducted via Zoom, interviews were audio-recorded and transcribed for analysis. Sample questions included: "*How much do you feel like you were really in the cleanroom during the VR-based microfabrication training?*" and "*After participating in VR-based microfabrication training, do you see yourself pursuing a career in the field of microfabrication?*"

F. Data analysis

Descriptive and inferential analyses evaluated changes in self-efficacy and interest. Paired-samples t-tests compared pre- and post-survey data for RQ1 and RQ2. A correlation matrix examined relationships among engagement, cognitive load, embodiment, and immersion (RQ3).

An explanatory sequential design integrated quantitative findings with qualitative insights (RQ1, RQ2, and RQ3). Interview transcripts underwent thematic analysis using Braun and Clarke's [26] six-step approach, identifying recurring themes. In this preliminary study, a single researcher conducted the coding, but all results—including codes and theme names—were reviewed and discussed with the full research team to reach consensus.

This mixed-methods approach ensured a comprehensive understanding of iVRLab's impact on students' affective factors and learning experiences.

III. RESULT

A. iVRLab's impact on self-efficacy

Descriptive statistics for self-efficacy are presented in Table I. A paired-samples t-test showed a significant increase in self-efficacy following the intervention, with the mean rising from 4.11 (SD = 0.60) to 4.37 (SD = 0.56), t(N−1) = −2.97, p = 0.0066.

TABLE I. DESCRIPTIVE STATISTICS FOR SELF-EFFICACY

Construct	Pre		Post	
	M	***SD***	***M***	***SD***
Self-efficacy	4.11	0.60	4.37	0.56

B. iVRLab's impact on interest

Table II presents descriptive statistics for interest, including two sub-constructs: Ideas about Careers and Ideas about Microelectronics. Paired-samples t-tests revealed slight increases for ideas about careers (M = 3.59 to M = 3.64, t = −0.89, p = 0.38) and ideas about microelectronics (M = 3.68 to M = 3.74, t = −0.72, p = 0.48), though neither change was statistically significant.

TABLE II. DESCRIPTIVE STATISTICS FOR INTEREST IN CAREERS AND MICROELECTRONICS

Construct	Pre		Post	
	M	***SD***	***M***	***SD***
Ideas about Careers	3.59	0.36	3.64	0.39
Ideas about Microelectronics	3.68	0.43	3.74	0.36

C. Correlated factors contributing to learning experiences

Table III summarizes correlations among cognitive load, engagement, embodiment, and immersion. Engagement was strongly correlated with embodiment (r = 0.91, p < 0.001) and moderately with immersion (r = 0.60, p < 0.01). Embodiment was positively correlated with immersion (r = 0.66, p < 0.001). Cognitive load showed no significant correlations with engagement (r = −0.03, p = 0.87), embodiment (r = −0.15, p = 0.47), or immersion (r = 0.08, p = 0.69).

TABLE III. THE CORRELATION MATRIX OF AFFECTIVE CONSTRUCTS (N = 26)

Variables	1	2	3	4
1. Cognitive Load	1.00			
2. Engagement	-0.03	1.00		
3. Embodiment	-0.15	0.91***	1.00	
4. Immersion	0.08	0.60**	0.66***	1.00

Note: ***p < 0.001, **p < 0.01, p < 0.05

D. Interview results

The interview results endorsed the iVRLab's effectiveness in delivering a positive learning experience, with key themes emerging:

1) Immersive and realistic learning environment: Six of seven participants highlighted the system's realism in bridging theory and practice. P9 stated, "*It kind of felt real you know like real*." P12 noted, "*Everything looked very realistic*."

2) Enhanced learning and retention: Five of seven participants credited interactivity with significantly improving learning and retention. P9 remarked, "*You can learn more during the VR than actually going in the lab*," while P1 emphasized, "*The hands-on aspect really helped; I learned way more than I thought I would.*"

3) Sense of control and autonomy: Five of seven participants appreciated the autonomy and pacing flexibility. P9 shared, "*It was set up in a way like the student was the one doing the moving and everything.*" P12 appreciated, "*You can learn at whatever pace you want.*"

4) Career interest and influence: Five of seven participants reported an increased interest in microfabrication careers. P9 stated, "*Learning about semiconductors definitely has me thinking about them,*" and P12 added, "*Doing this, I was able to actually understand it…now it's something I more likely would like to do.*"

5) Challenges and frustrations: Four of seven participants reported minor technical issues like lag and headset discomfort. P9 noted, "*It was kind of slow,*" while P1 mentioned, "*Sometimes the headset was a little uncomfortable*."

6) Future of VR in education: Six of seven participants expressed optimism about VR's educational potential. P9 said, "*It's going to play a big role in student learning for sure,*" and P12 emphasized, "*This will allow more people to learn about the basics*."

IV. Discussion and Implications

This study examined iVRLab's effects on college students' affective factors—self-efficacy, career interest in science and technology, and microelectronics interest—within an informal learning setting. Findings demonstrate iVRLab's effectiveness in strengthening self-efficacy, maintaining interest, and fostering positive learning experiences, underscoring VR's potential in STEM education beyond traditional classroom environments.

A. RQ1: How does the iVRLab intervention impact students' self-efficacy in microelectronics before and after participation?

The iVRLab significantly improved self-efficacy, with mean scores increasing from 4.11 to 4.37 ($p = 0.0066$). Realistic simulations, interactive feedback, and hands-on practice in a risk-free environment helped participants master complex tasks, aligning with prior research on VR's role in boosting confidence [8], [11]. Participants expressed heightened assurance, with one noting, "*I feel like I can handle the real cleanroom now*."

B. RQ2: How does the iVRLab intervention impact students' interest in microelectronics before and after participation?

Quantitative results showed only slight increases in interest—Ideas about Careers ($M = 3.59$ to $M = 3.64$) and Ideas about Microelectronics ($M = 3.68$ to $M = 3.74$)—on a 4-point scale, which were not statistically significant. One likely explanation is the voluntary nature of the camp, where participants may have already had a strong interest in microelectronics. Their high baseline interest (3.59 and 3.68, respectively) suggest limited room for measurable growth. It is also worth noting that the consistently high interest levels observed in both pre- and post- surveys suggest that the overall camp experience—including our iVRLab activity—effectively supported and sustained students' interest in microelectronics. Despite this ceiling effect [27], qualitative data confirmed a strong positive impact on curiosity and aspirations; for example, one participant remarked, "*I hadn't thought much about microelectronics before, but now I want to learn more.*" While the short intervention helped maintain engagement, more extended VR use could further deepen career interest.

C. RQ3: How do students perceive other aspects of their learning experiences, such as engagement, immersion, and cognitive load, during the iVRLab intervention?

Engagement, embodiment, and immersion strongly correlated, highlighting their interdependent role in fostering effective learning. Engagement correlated strongly with embodiment ($r = 0.91$, $p < 0.001$) and moderately with immersion ($r = 0.60, p < 0.01$), supporting Mayer's CTML [17], which highlights learners' limited cognitive capacity for processing information from auditory and visual channels simultaneously. The iVRLab's immersive VR environment effectively managed cognitive load by integrating multimodal instructional design, allowing students to actively engage in tasks without overwhelming their cognitive resources. Cognitive load showed no significant correlations, indicating effective management of cognitive demands.

Participants highlighted the iVRLab's immersive environment, interactive features, and autonomy in learning. Real-time feedback and task repetition were praised for reinforcing understanding, as one noted, "*I remember the steps better because I actually did them*." These findings emphasize the value of active, embodied learning in VR settings [8].

D. Implications for microelectronics education

The iVRLab demonstrates VR's potential to bridge theoretical learning and practical application, especially in resource-intensive fields like microelectronics. Additionally, iVRLab practice can reduce logistical demands, conserve human resources, increase safety, and make repeated practice more accessible compared to traditional methods [8], [9]. By fostering self-efficacy and technical skills, VR effectively prepares students for real-world challenges. Integrating VR into broader curricula and informal learning environment with repeated engagements could further enhance aspirations and outcomes.

Expanding VR learning systems to improve engagement, embodiment, and immersion fosters retention and active participation. Leveraging VR in STEM education addresses accessibility and inclusion challenges, engages diverse learners, and meets industry demands. By incorporating VR's immersive capabilities, educators can equip future professionals with the skills to excel in evolving technological landscapes.

V. Limitations and Future Directions

This study's limitations include a small sample size, brief session duration, and a focus on photolithography, restricting generalizability. Future research should involve larger, diverse cohorts, longitudinal studies, and explore VR applications in areas like circuit design. Stratifying learners by prior knowledge can provide deeper insights into VR's effectiveness.

The lack of a control group limits isolating VR's impact, emphasizing the need for comparative studies with traditional labs. Addressing technical issues like lag and headset discomfort is essential. Expanding VR to other STEM fields, integrating it into curricula, and studying its impact on underrepresented groups can enhance diversity and optimize STEM education.

VI. Conclusion

The findings demonstrate iVRLab's potential to enhance self-efficacy and sustain interest in microelectronics education as an informal learning tool. By simulating photolithography in a realistic virtual cleanroom, it boosted confidence in performing complex tasks, aligning with research on VR's role in skill development. Although interest gains were modest, likely due to already high baseline levels, qualitative feedback indicated heightened curiosity about microfabrication careers.

Correlations among engagement, immersion, and embodiment suggest the potential value of active, immersive VR systems in enhancing students' learning experiences. Addressing limitations such as small sample size and brief sessions is crucial for broader applicability. Future research should integrate VR into curricula, expand its STEM applications, and refine systems to meet evolving microelectronics industry demands.

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